

A simple and inexpensive system for monitoring jaw movements in ambulatory humans

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Abstract

A simple and inexpensive method for recording vertical movements of the human mandible relative to the maxilla is presented. Measurements are made from accelerometers and a Hall-effect device temporarily glued to the upper and lower anterior teeth. The accelerometer signals are integrated once to give velocity and a second time to give position. Movements of the mandible relative to the maxilla are obtained by integrating the difference between the two accelerometer signals. The (relative) velocity and position records derived in this way are linear, but subject to drift when the jaw is stationary. Steady mandibular position is obtained from the Hall-effect system, but this signal must be corrected for its inherent non-linearity. This device can record rapid movements of the mandible even when the head is unrestrained, and interferes minimally with normal jaw movements. © 2002 Elsevier Science Ltd. All rights reserved.

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1. Introduction

A number of methods have been used to measure vertical jaw position in humans (Griffin and Malor, 1975; van Rensburg, 1975; Hannam et al., 1980; Miles and Wilkinson, 1982; Ostry and Flanagan, 1989; Wilding and Lewin, 1991; Plesh et al., 1993; Hayashi et al., 1994; Naeije et al., 1995; Gallo et al., 2000). In most cases, the measurement is made using some form of transmitter that is glued to a mandibular tooth while a sensor is attached to the head. Many of the systems that have been described are relatively bulky, heavy, complicated to attach, or require the head to be fixed, and are expensive (e.g., Griffin and Malor, 1975; van Rensburg, 1975; Hayashi et al., 1994; Miles and Wilkinson, 1982). While some of these systems measure jaw movements in three dimensions, the important measurement in most studies of jaw reflexes is the vertical position of the mandible relative to the maxilla (e.g., Hannam et al., 1968; Miles and Wilkinson, 1982; Miles et al., 1995; Wang et al., 2000).

Here we present a simple, inexpensive method to measure vertical movements of the mandible in relation to the maxilla, which is suitable for situations where the head is unrestrained and when rapid jaw movements are to be measured. This system is intended to measure only over the range of the rotational component of vertical human jaw movements, and not over the rotation plus translation component.

2. Materials and method

Informed, signed consent was obtained from 8 volunteers, and the experiments were carried out with the approval of the Human Research Ethics Committee of Adelaide University.

Two methods were used in parallel to measure jaw position and movements. The first was based on records of the acceleration of the mandible relative to the maxilla. Single-axis piezoelectric accelerometers with an output of 400 mV/m s^2 (model ADXL105Q, Analog Devices, Norwood, MA, USA) were chosen for their size ($7 \times 9 \times 4 \text{ mm}^3$), low inertial mass ($<1 \text{ g}$) and broad-band linearity (DC to 5 kHz). The cost of the accelerometers was approximately \$US50 each and the

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Hall-effect system approximately \$US40. The accelerometers require a ± 5 VDC power supply, which was provided by batteries capable of supplying 50 mA.

Velocity and position signals were derived from the accelerometer outputs. The electrical output signals of the two accelerometers were summed electronically in a precision differential amplifier to give the acceleration of the mandible relative to the maxilla. The resulting differential acceleration record was digitised at a 5 kHz sampling rate with a CED laboratory interface (1401plus, CED Ltd., Cambridge, UK), mathematically integrated once to give the velocity of the mandible relative to the maxilla, then integrated a second time to give relative jaw position. Integration was performed with software supplied with the laboratory interface (CED Signal version 1.85) but could equally be achieved with integrating amplifiers. The operation of the system was checked by asking subjects to shake and nod their head whilst clenching their teeth. In this situation there are large peaks in the acceleration records of the maxilla and mandible but their relative acceleration is zero.

The major shortcoming of the accelerometer-based system is that it does not give stable records of static jaw position. To overcome this, a second transducer system was used, based on a Hall-effect device for measuring small distances. This consisted of a powerful rare-earth magnet 5 mm diameter and 2 mm thick whose field was detected by a Hall-effect magnetic sensor (part # 304-267, RS Components, Corby, UK). The effective working range of this device was 1–25 mm. The Hall-effect device gives a constant output when the separation between the magnet and the sensor is constant, and is calibrated by asking the subjects to grip spacers of known thickness between their teeth. Its non-linear output was compensated by re-scaling by a curve-fitting program implemented off-line in a spreadsheet.

One accelerometer and the magnet were glued to a bracket formed from a double thickness of 1.5 mm lead-free and resin-free solder wire that enabled it to be adapted to lip shape, dental overbite, etc. This bracket was then glued to the labial surface of an upper central incisor tooth with fast-curing, glass ionomer dental cement (GC Fuji IX GP, GC Corporation, Tokyo, Japan) as shown in Fig. 1.

This cement is normally applied to an acid-etched enamel surface, but the mechanical strength of the bond obtained without etching was appropriate for the present purpose. The bracket was easily removed from the tooth at the end of the experiment and small specks of residual cement were cleaned from the tooth with a hand-held dental scaler. The second accelerometer and the Hall-effect sensor were mounted on a similar bracket, which was glued to the labial surface of the lower incisor tooth on the same side. Care was taken to keep this bracket from contact with the upper bracket and teeth when the subject clenched.

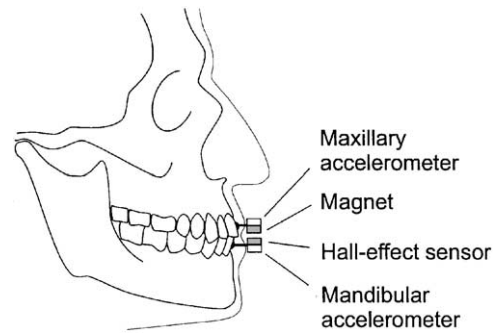


Fig. 1. Mounting of transducers to teeth. A flexible metal bracket carrying one transducer array is glued to an upper incisor tooth; another bracket carrying the second array is mounted directly below this on a lower incisor tooth. The canine teeth can also be used. The brackets are bent to enable them to pass between the lips to keep the transducers outside the mouth and optimally aligned to each other. Given appropriate waterproofing, it would also be possible to have the transducers inside the mouth, attached to the buccal surfaces of posterior teeth. Fine wires (not shown) carry the +5 V power to the transducers and the output signals to the data acquisition system.

The accelerometers were connected by fine, flexible ribbon cables to instrumentation amplifiers to match gains and allow offset adjustment of static acceleration (INA 125PA, Burr Brown, Tucson, USA). Other fine wires brought the ± 5 VDC power for the accelerometer and the Hall-effect transducer. The lower and upper sensor assemblies weighed 3 and 2.5 g, respectively.

To check and calibrate the performance of the system, the devices were mounted on an apparatus used in reflex studies to apply precise stretches to the jaw-closing muscles (Miles et al., 1993). This “muscle-stretcher” system simulates the rotational component of movement of the human temporomandibular joint. That is, the lever arm on which the mandibular accelerometer was mounted was co-axial with the temporomandibular joint, and the bite plates on which the devices were mounted correspond with the positions of the incisor teeth (Fig. 1 in Miles et al., 1993). With this device, the transducers were moved known distances relative to each other at known angular accelerations under servo control.

In a series of preliminary experiments, the mandibular movement was measured in 8 human subjects who jumped down from a 100 mm step and landed on one heel. The movement signals recorded during 8 hops were averaged, using the rapid deceleration of the maxilla on landing as a trigger.

3. Results

Fig. 2 shows the results of bench-testing the system on the servo-controlled displacement device. The position record obtained by integrating the accelerometer output

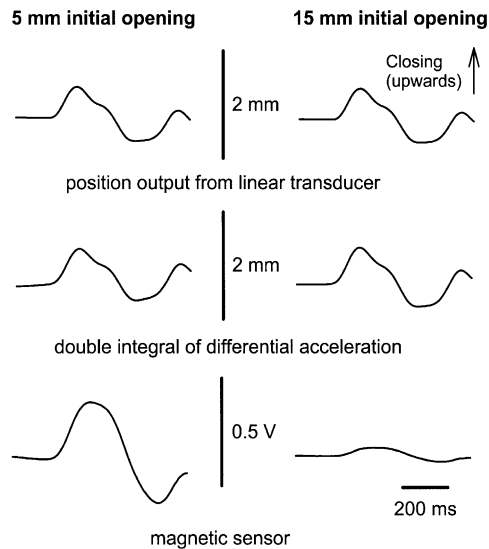


Fig. 2. Comparison of the position record derived in a bench test from a precision linear output transducer (uppermost traces), double integration of the mandibular accelerometer (middle traces), and from the Hall-effect device (lowermost traces). A servo-controlled displacement device moved the mandibular transducer in relation to the maxillary transducer at two initial separations (simulating two initial vertical separations of the incisor teeth). The movement consisted of a 0.5 mm upward (“closing”) then downward (“opening”) movement, before returning to the initial position. The vertical velocities were $\approx 50 \text{ mm s}^{-1}$. The left column shows the data obtained at the first “jaw opening” of 5 mm, and the right column the data obtained at 15 mm separation. The non-linear input–output properties of the Hall-effect device resulted in a signal whose gain is reduced about 10-fold at 15 mm compared with 5 mm, while the double-integrated output from the accelerometers is constant irrespective of starting position. All signals shown are the averages of 50 trials.

corresponded closely with the output of the linear displacement transducer on the servomotor over accelerations ranging from 0.5 to 40 m s^{-2} . Note, however, that the gain of the Hall-effect device is reduced about 10-fold when tested at a separation (“jaw opening”) of 15 mm, compared with the gain when the separation was 5 mm.

Vigorous head-shaking with the teeth clenched resulted in a zero difference signal in 8 subjects tested, indicating no relative acceleration (or movement) between upper and lower jaw. A minor misalignment of the upper and lower accelerometers would cause a difference in sensitivity, and thus a small non-zero signal in the difference signal. This can be corrected by adjusting the gain of the other sensor.

The output of the Hall-effect device was calibrated and corrected for non-linearity using a curve-fitting program implemented in a spreadsheet, off-line. Fig. 1 shows that this “linearised” output also correlated well with the linear displacement transducer.

Fig. 3 shows representative data obtained in an experiment in which the jaw position was monitored

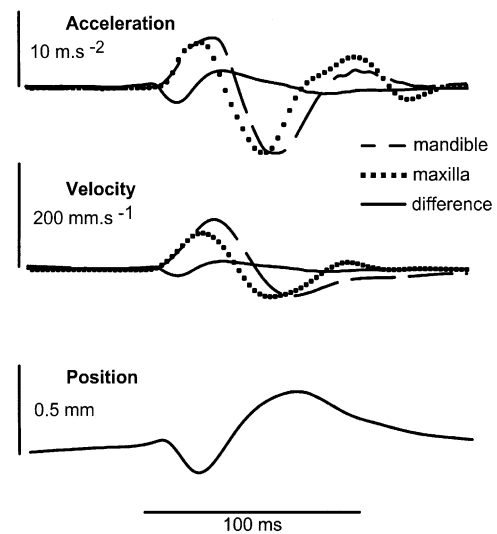


Fig. 3. Jaw movements measured during landing on one heel while jumping from a height of 100 mm. The resulting sudden deceleration upon landing caused the mandible to move downwards with respect to the maxilla. The uppermost panel shows the output from accelerometers mounted on the maxillary and mandibular teeth and the difference signal. The result of integrating these signals once to give maxillary and mandibular velocity is shown in the middle panel. The lowermost panel shows the result of a second integration of the differential acceleration signal to give vertical jaw position relative to the maxilla. Thus, heel landing caused the mandible to move about 0.2 mm downwards relative to the maxilla. All signals are averages of 8 trials, triggered by the rapid deceleration of the maxilla on landing.

while a subject jumped down from a 100 mm step and landed on his heel. The accelerometer signals shown in the upper panel are the ensemble averages of 8 trials. The difference trace (relative acceleration) was obtained by averaging the mathematical difference between maxillary and mandibular accelerometer records in each trial. The result of integrating these three signals to give, respectively, the velocities of the maxilla and mandible, and the velocity of the mandible relative to the maxilla are shown in the middle panel. The lowermost panel shows the result of the second integration of the difference signal to give the vertical *position* of the mandible relative to the maxilla. Landing caused the head to tilt forwards, which initially moved the mandible slightly closer to the maxilla before it moved downwards away from the maxilla. It then overshot upwards (presumably as the result of reflex activation of the jaw-closer muscles), but the teeth did not touch: touching of the teeth would appear as a sharp deceleration of the mandible in the uppermost trace, and as a flat peak in the mandibular position trace.

The accelerometer system was relatively non-intrusive and was tolerated well by subjects who wore it during manoeuvres such as jumping, hopping and treadmill-running.

The response of the accelerometers is linear up to 5 kHz, which exceeds by several orders of magnitude the highest frequency at which the mandible moves under physiological conditions (van Eijden, 2000).

4. Discussion

The teeth are rigidly fixed in the bone: hence, direct mechanical coupling of sensors to the lower and upper teeth allows the movements of the mandible relative to the maxilla to be estimated accurately.

The accelerometers and magnet systems each have advantages and disadvantages. The accelerometers give an acceleration signal that is valuable in, for example, reflex studies in which detecting the onset of jaw movements is critical (Hannam et al., 1968; Miles and Wilkinson, 1982; Miles et al., 1995; Wang et al., 2000). Jaw velocity can then be measured accurately by integrating the acceleration signal. Measurement of position of the jaw relative to the maxilla is more difficult. Double integration of acceleration gives a signal that is linearly related to position over a wide range of jaw openings; however, the drift inherent in the integration process prevents this method being used to measure a constant jaw position, or a very slow-moving jaw. The magnetic device gives a signal proportional to jaw position even when the jaw is not moving, but the relationship is non-linear and its resolution declines as the jaw opening increases. In practice, using both signals gives information about jaw position and movement over the whole range of jaw positions and velocities.

This system is very cheap to implement, and is simple and non-invasive. It has been shown in a series of hopping and treadmill-running experiments (e.g., Fig. 3) to be reliable and suitable for monitoring various parameters of natural jaw movements in the laboratory and in the field. Signals from the sensors are low noise and require minimal conditioning prior to storage on portable analog or digital recorders.

The performance of the system in measuring vertical jaw movements is at least equal to that of commercially available optical tracking systems. One such system (Optotrak 3020, Northern Digital Inc., Canada) has an accuracy of 0.1–0.3 mm but is unavailable as a real-time signal. The systems based on magnet-sensing give real-time outputs but have coarser resolutions, e.g., 0.3 mm (Gerstner and Fehrman, 1999; Gerstner and Kinra, 1999), and 0.5–4 mm (Hannam et al., 1980; Wilding and Lewin, 1991).

The accelerometer approach could be used to monitor lateral as well as vertical jaw movements using dual axis accelerometers of the same size (e.g., ADXL205Q Analog Devices, Norwood, MA, USA). The Hall-effect method would require two magnets and two sensors but

would require careful alignment to avoid problems arising from the interaction of the two magnetic fields.

This system could be also adapted for use in experimental animals. No modification would be necessary for use in anaesthetised animals. In free-ranging animals, appropriately waterproofed transducers could be glued directly to the buccal surface of posterior teeth after etching of the enamel to give the cement a stronger bond, and connected to a data logger.

This system provides a simple, low-cost solution to the problem of jaw-movement monitoring in biomechanical and physiological studies.

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